



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

GAZIPUR-1704, DHAKA, BANGLADESH

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

PROJECT REPORT

EEE 4518

ELECTRICAL AND ELECTRONIC WORKSHOP

***PROJECT TITLE:* Enhancement of Battery-Powered Rickshaw.**

SUBMITTED BY:

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Table of Contents

<u>1. Introduction</u>	3
<u>2. Objectives</u>	4
<u>3. Required Components</u>	5
<u>4. Circuit Diagram</u>	5
<u>4.1 Explanation:</u>	6
<u>5. Features</u>	6
<u>6. Working Principle</u>	7
<u>6.1 Flowchart:</u>	7
<u>6.2 Explanation:</u>	8
<u>7. Code</u>	8
<u>8. Hardware Demonstration</u>	10
<u>9. Problems Faced</u>	11
<u>10. Conclusion</u>	11
<u>11. References</u>	12
<u>12. Contributions</u>	13
<u>13. CO PO covered</u>	19

1. Introduction

➤ **Problem Statement**

-Battery-run rickshaws are the cause of a good number of problems in the urban areas of our country. The major issues that we will be addressing are:

1. Unnecessary speeding of battery rickshaws.
2. Collisions that lead to minor or severe injuries and life threats
3. Excessive use of electricity for charging the battery-run rickshaws
4. Lack of safety measurements in the rickshaws
5. Absence of an information system in the rickshaw.

Battery-run rickshaws are most criticized for their poor safety measurements and functionality. The major issue is that these rickshaws do not have any safety options to control overspeeding. The speeds these rickshaws can gain are often too hard to control with the brakes they are equipped with. These lead to collisions with other vehicles on the road and serious injuries, sometimes even death. The inclusion of speed sensors and automatic response systems for overspeeding can resolve this issue to an extent.

Another problem that the battery-run rickshaws create is the electricity they consume for charging. A study conducted in 2023 shows that the battery-run rickshaws consume up to 450 MW of electricity every day. However, this number might be even larger as there are many illegal connections used for charging these rickshaws. To address this issue, the use of renewable energy systems and the addition of a hybrid mechanism is proposed.

➤ **Existing Solution**

For this case , usually traditional battery rickshaws use illegal line to change their batteries. Normally in the accident prone areas , rickshaw-pullers manually sense the speed and obstacles as there is no established technologies to detect over speeding and obstacles .In our country , manual brakes are used when any chance of collision arises .Indicator lamps and headlamps are not used that much but in the modern rickshaws headlights are used now a days .Led display is used rarely in the modern rickshaws. But in the traditional rickshaws , there isn't any technology available to show necessary information.

➤ **Proposed Solution**

-To solve the mentioned problem statements above, we aim to use several sensors and a hybrid mechanism in battery-run rickshaws accompanied by the use of renewable sources to cater to power saving. The proposed modifications are:

1. Solar panel installation on the top of rickshaw hoods
2. Speed sensors to detect the speed of the rickshaw
3. Obstacle detector to be installed at the front portion
4. A system that can respond to overspeed and reduce chances of collisions by controlling the power supply to motor.
5. Addition of indicator lamps and headlamps
6. Information system (LED display) to show necessary data (speed, charge, etc.)

The prototype we aim to make will comprise of the following modules:

- Arduino
- Solar panel
- Speed measuring module
- Ultrasonic SONAR sensor (HC-SR04)
- Dual Motor drivers
- LCD module
- LED
- An alternator

2. Objectives

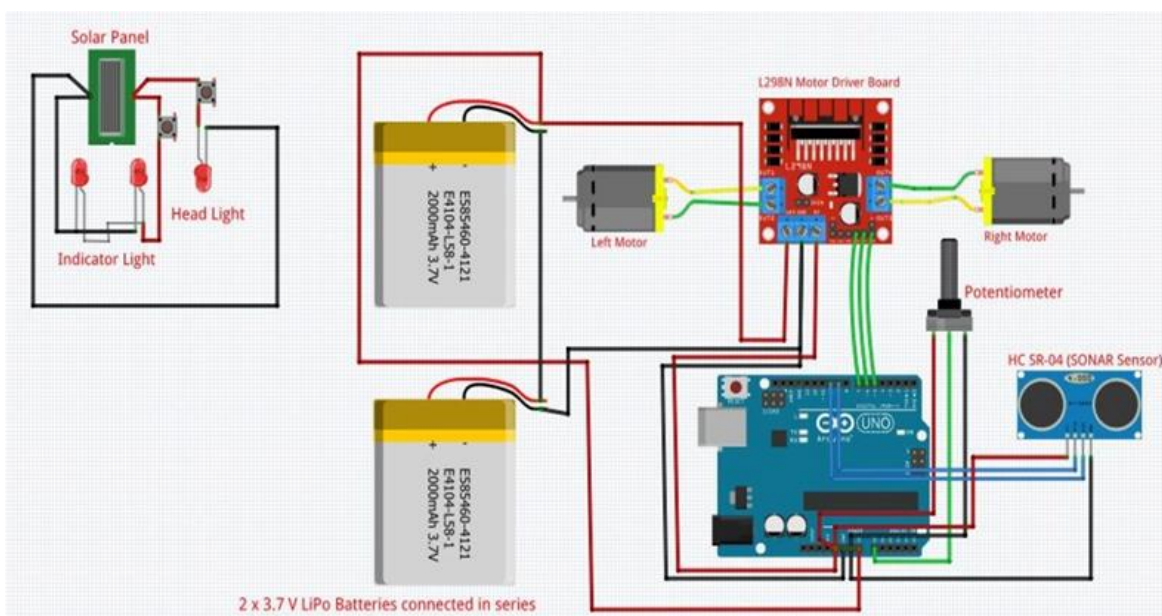
Team **murubbEEE** intends to work on enhancing battery-run rickshaws in Bangladesh. Our aim is to ensure safety, increase the efficiency and sustainability of battery-run rickshaws . The proposed enhancement of the systems we want to go through is :

- Reducing reliance on grid electricity by incorporating solar panels.
- Ensuring safety by bringing in speed sensors, obstacle detector and automatic speed regulation system
- Installing functions like indicator lamps, headlamps and LED display for ensuring safety and enhancing data monitoring
- Utilizing **Arduino-based automation**, motor drivers for optimizing the control system and an LCD module for providing efficient operation and feedback.

3. Required Components

Sl No.	Component Name	Value/Model	Quantity	Price (BDT)
1	Aurduino – UNO	R3	1	995/-
2	Motor Driver	L298N	1	120/-
3	Potentiometer	10K	2	30/-
4	Sonar sensor	HC SR-04	1	60/-
5	LiPo Battery	11 V	1	1600/-
6	Motor	25GA 370,12V gear	2	800/-
7	Solar Panel	RYO 474	1	160/-
8	LED		1	10/-
9	Display	LCD 16*2 Blue Backlight	1	155/-
10	Adapter Module	I2C LCD	1	90/-
11	Jumper Wires + HEXA	-	1 set+2	100+160/-
12	Soldering Kit		1 set	400/-
Total				4680/-

4. Circuit Diagram



4.1 Explanation:

The usage and significance of the components used in this project are discussed below:

- **Aurduino UNO** :- It works as central controller device. It reads datas from sensors and adjust various functions. From speed sensor , it reads the command and thus adjusts motor speed . From sonar sensor's input, it triggers an automatic braking response . Also aurduino balances the operations of solar panel , LED and other integrated devices .
- **Motor Driver** :- It controls motor's speed and direction and further ensuring precise control . It also cntrols the power supply to the rickshaw by observing the speed and obstacle .
- **Potentiometer** :- It helps to regulate the speed of the rickshaw. Acting as a speed control device, it also adjusts voltage for motor control.
- **Sonar Sensor** :- It detects obstacles to prevent collisions. Measuring the distance between rickshaw and objects ,it sends alerts and triggers automatic braking if the object is likely to collide in a short time.
- **Motor** :- It acts as the primary propulsion system . It works by converting electrical energy into mechanical motion, thus optimizing efficiency.
- **Solar panel** :-It's a device which converts sunlight into the electrical energy to charge the battery.It also reduces the dependency on power grid and also saves electricity costs .
- **LED**:- It ensures safety by making the rickshaw more noticeable . Actually LED is used as indicator or headlamps.
- **LiPo Battery** :- It provides power to the electronic components as well as to the rickshaw. At the same time, it also increases battery health and reduces charging time.
- **Display** :- It provides instant feedback to the driver by showing different real time data like speed , battery status and alerts .
- **Adater Module** :- It helps in efficient power distribution between the electronic components.
- **Jumper wires , Hexa and Soldering Kits** :- To connect the electronic components jumper wires were used .To fix the components with board Hexa and soldering kits were used .

5.Features

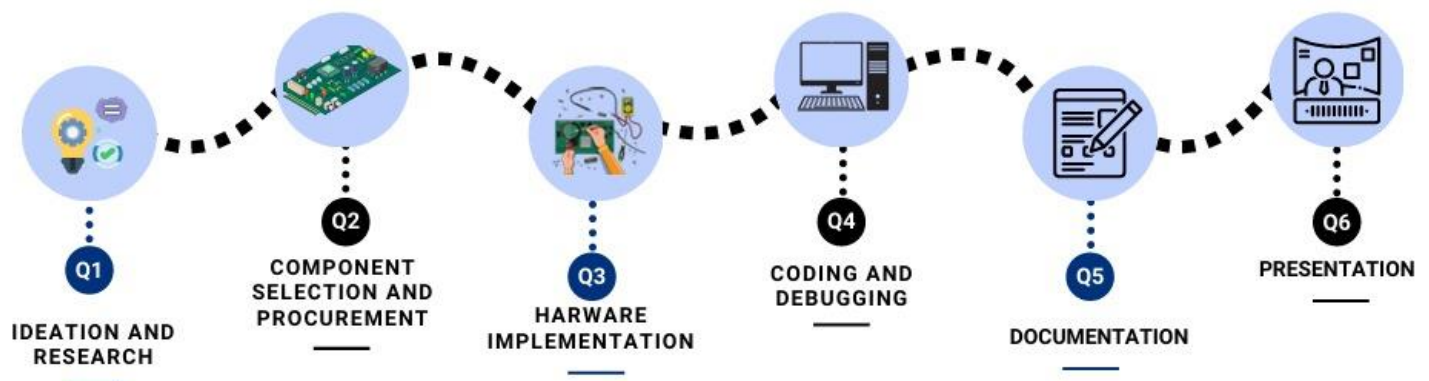
Team *murubbEEE* provides some indistinctive features for the enhancing performance of battery-run rickshaw model. The features are given below:

- Automatic Speed detection system (potentiometer)
- Obstacle detection system (sonar sensor)
- Energy saving system (solar panel)
- Digital data portraying system (LCD display)
- Enhancing safety measurements (LED)

6. Working Principle

The components used in the model have different functions in enhancing the performance. The ARDUINO UNO receives data from the sensors and works accordingly. Speed sensor is used to detect speed and potentiometer is used in controlling the speed. The SONAR sensor is used to detect objects and the automatic braking system serves in avoiding sudden collisions. LED headlamps and the indicator lamps turn on automatically when it is needed such as: low-light conditions and sudden turning/braking. The monitor used in the display system keeps the driver updated about the obstacle alerts, battery charge level and speed. The solar panel installed over the hoods capture sunlight and convert it into electrical energy. It reduces dependency on the external charging and reduces pressure on the electrical grid line.

6.1 Flowchart:



6.2 Explanation:

Here's a summary of the steps involved in using the enhanced Battery-run rickshaw system:

At first we used two motors for this project . To control and run the motor , we connected the motors with Motor driver (L298N) . As motor driver cant work by itself and it requires command to work , it was connected to a micro-controller circuit named Aurduino UNO R3 . Then necessary codes were given as the input of Aurduino . The Aurduino was connected to other electronic components . To control motor speed , we used the potentiometer and we used sonar HC-SR04 to detect obstacles . This works as an automatic braking system which can detect obstacles within 20cm . The LCD display that we used has 16 pins . To reduce the pin numbers we used an extra module named I2c which converts 16 pins into 4 pins . To provide power to all the components , we used a LiPo Battery containing 12V . Remaining components (Indicator, headlamps) were powered by a solar panel which we intend to put on the hood of the rickshaw . We aimed to reduce extra energy losses .

7. Code

```
#include <LiquidCrystal_I2C.h>
#include <Wire.h>

// HC-SR04 Ultrasonic Sensor pins
#define trigPin 9
#define echoPin 10

// Motor A pins (Left motor)
#define motorEnableA 5
#define motorIn1 6
#define motorIn2 7

// Motor B pins (Right motor)
#define motorEnableB 11
#define motorIn3 8
#define motorIn4 12

// Potentiometer pin
#define potPin A0

// Distance limit to stop the motors
const int stopDistance = 20; // Adjust as needed (in cm)

// Initialize the LCD with I2C address 0x27, 16 columns, and 2 rows
LiquidCrystal_I2C lcd(0x27, 16, 2);

long duration;
int distance;

void setup() {
  // Initialize Serial Monitor
  Serial.begin(9600);
```

Project name by team name

```

// Set up motor pins
pinMode(motorEnableA, OUTPUT);
pinMode(motorIn1, OUTPUT);
pinMode(motorIn2, OUTPUT);

pinMode(motorEnableB, OUTPUT);
pinMode(motorIn3, OUTPUT);
pinMode(motorIn4, OUTPUT);

// Set up ultrasonic sensor pins
pinMode(trigPin, OUTPUT);
pinMode(echoPin, INPUT);

// Initialize the LCD
lcd.init();    // Initialize the LCD
lcd.backlight(); // Turn on the LCD backlight
lcd.print("Motor Speed:");
}

void loop() {
    // Ultrasonic sensor reading
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2; // Convert to cm

    // Read potentiometer for speed control
    int potValue = analogRead(potPin);
    int motorSpeed = map(potValue, 0, 1023, 0, 178); // 70% of max speed

    // Display speed on LCD
    lcd.setCursor(0, 1);
    lcd.print("Speed: ");
    lcd.print(motorSpeed);
    lcd.print("  "); // Clear extra characters if overwritten

    // Motor control based on distance
    if (distance >= stopDistance) {
        // Run motors if no object within stop distance
        analogWrite(motorEnableA, motorSpeed);
        analogWrite(motorEnableB, motorSpeed);

        // Set direction (forward)

```

```

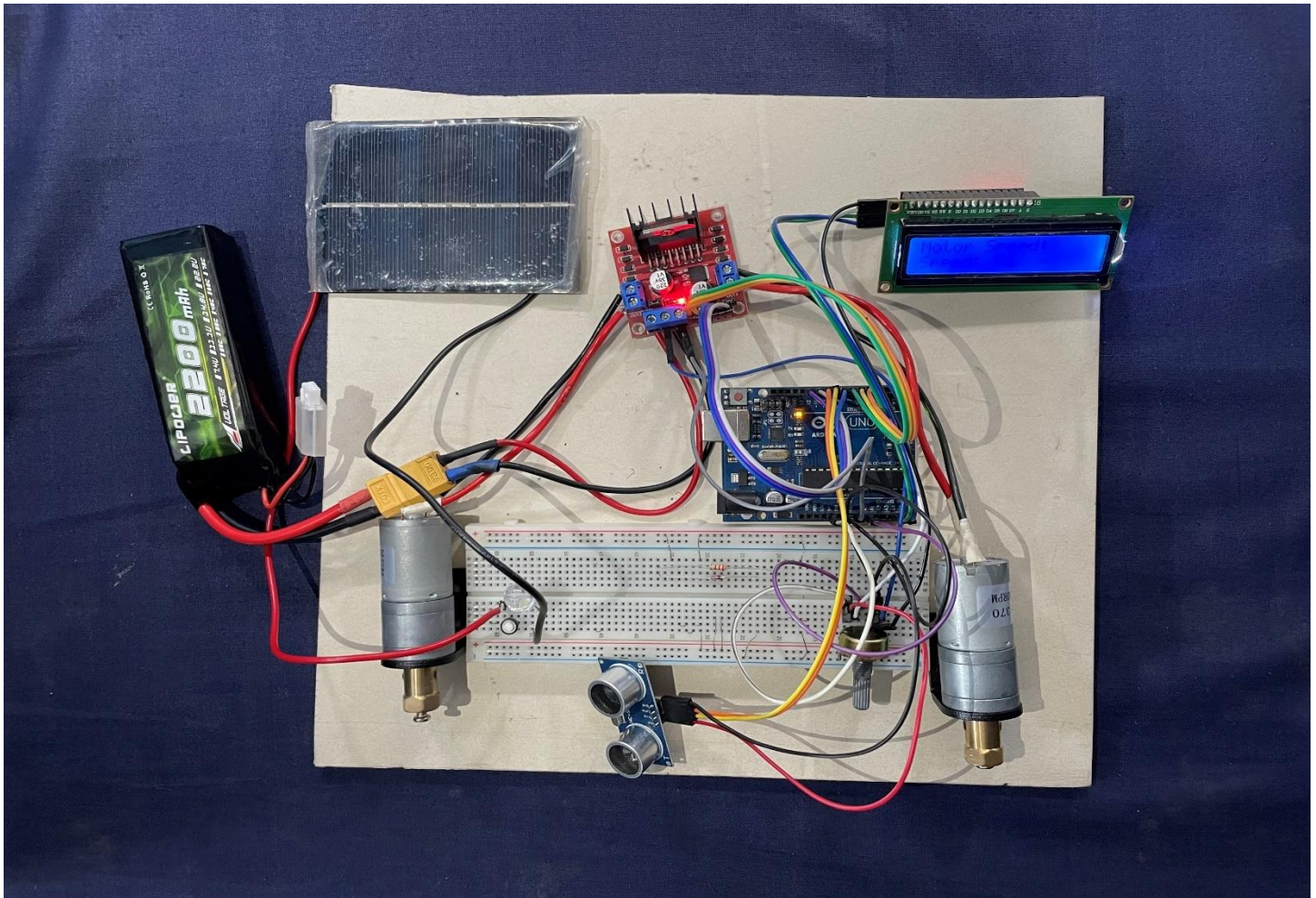
digitalWrite(motorIn1, HIGH);
digitalWrite(motorIn2, LOW);
digitalWrite(motorIn3, HIGH);
digitalWrite(motorIn4, LOW);
} else {
  // Stop motors if object is within stop distance
  analogWrite(motorEnableA, 0);
  analogWrite(motorEnableB, 0);
}

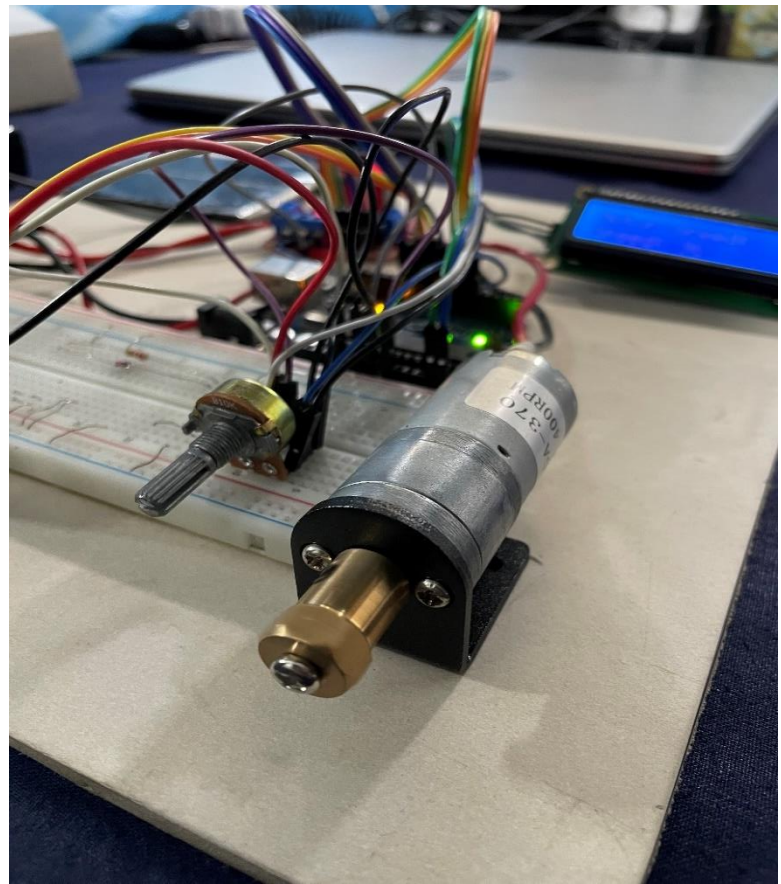
delay(100); // Short delay for stability
}

```

8. Hardware Demonstration

The figure below shows the finished HW part





Video link of prototype: <https://www.youtube.com/shorts/Rr9jDDKW8tM>

9. Problems Faced

We encountered some challenges while generating the ARDUINO code. We faced an issue with the **SONAR sensor** while implementing the hardware, as it failed to detect obstacles properly due to internal hardware problems. It might need multiple attempts to function correctly unless held in a specific position. Additionally, the **LED** we initially used did not respond properly. Initially, we suspected an issue with the **solar panel**. But after replacing the **LCD**, we confirmed that the solar panel was functioning correctly, and the problem was solely with the previous **LCD**.

10. Conclusion

In this project, we utilized the functions of a Battery-run rickshaw. This Enhanced Battery-powered rickshaw model represents a sophisticated integration of sensor technologies, digital data-monitoring systems & energy saving mechanism to ensure safety & increase efficiency. In this model, we used SONAR sensor, Solar panel, LCD, LED, MOTOR driver and Potentiometer. Our intention to ensure safety while enhancing performance and saving energy were accomplished by the above mentioned components. This model is also beneficial for natural and economical perspectives as it reduces Carbon footprint and increases durability. As we are saving energy by using the Solar Panel, it is also helping us not to rely heavily on power grids for electricity.

Overall, the project is a successful implementation of solar energy and sensors ensuring proper safety.

11. References

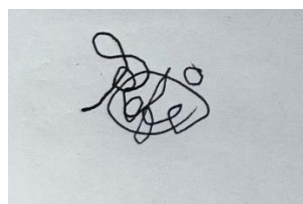
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2. ENERGY CONSUMPTION BY BATTERY OPERATED AUTO-RICKSHAWS. (2013). *DAFFODIL INTERNATIONAL UNIVERSITY JOURNAL OF SCIENCE AND TECHNOLOGY*, 8(2), 71.
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3. Issue-I, T. A. (2022, February 2). *Impact of ban on battery-run rickshaws in districts*. The Financial Express. <https://thefinancialexpress.com.bd/views/columns/impact-of-ban-on-battery-run-rickshaws-in-districts-1643812426>
4. https://www.tinkercad.com/things/9a0EzJV0BrF/editel?sharecode=sQkeSWPK3nWY5nMIFGdWmH7SoYoP_1VOd5YGY4Q7PtE

12 Contributions

a. Contribution of team member 1

AR-RAFI ISHRAQ

I researched on the component selection and found out the necessary components for our project with my team mate number 2,3 & 4. After fixing the component list, I went to buy the necessary components with my team number 2,4,5 & 6 from Patuatuli , Puran Dhaka. Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I set up the components. I searched up the codes for individual Modules and modified the code according to our project. While coding, at first, the codes weren't working and then necessary debugging was done to run the code properly. . Necessary debugging in the hardware portion was also done by me & my team member number 2.

A handwritten signature in black ink, appearing to be 'Ar-Rafi Ishraq', written on a light blue background.

Signature

b. Contribution of team member 2

ISHRAK ANAN

I researched on the component selection and found out the necessary components for our project with my team mate number 1,3 & 4. After fixing the component list, I went to buy the necessary components with my team number 1,4,5 & 6 from Patuatuli, Puran Dhaka.

Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I contributed in wiring and soldering with my team mate number 3,4,5 & 6. Necessary debugging in the hardware portion was also done by me & my team member number 1. I made the whole presentation slide with the help of my team member number 3.



Signature

c. Contribution of team member 3

TAUFEEQ HASSAN OMIO

I researched on the component selection and found out the necessary components for our project with my team mate number 1,2 & 4. I also researched about the project feasibility and impacts of the projects while writing the project proposals with my team mate number 4 & 6. Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I contributed in wiring and soldering with my team mate number 2,4,5 & 6. Then I, with the help of my team mate number 4, wrote the whole report. I also contributed in making the presentation slide with my team mate number 2.

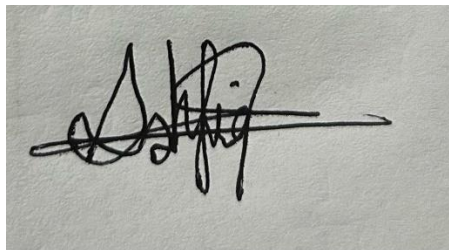
A photograph of a piece of grey paper with the word 'Omio' written in black ink in a cursive, handwritten style.

Signature

d. Contribution of team member 4

ASHFIQ UL RAHMAN

I researched on the component selection and found out the necessary components for our project with my team mate number 1,2 & 3. After fixing the component list, I went to buy the necessary components with my team number 1,2,5 & 6 from Patuatuli, Puran Dhaka. I also researched about the project feasibility and impacts of the projects while writing the project proposals with my team mate number 3 & 5. Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I contributed in wiring and soldering with my team mate number 2,3,5 & 6. Then I, with the help of my team mate number 3, wrote the whole report.

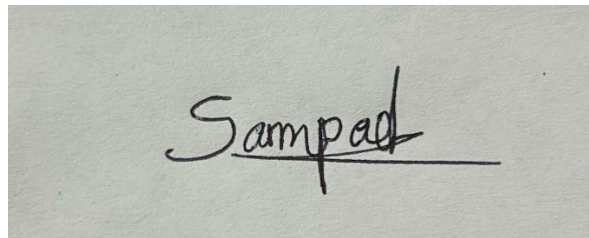
A handwritten signature in black ink on a light-colored background. The signature is stylized, with a large, looped 'A' and a long horizontal stroke extending to the right.

Signature

e. Contribution of team member 5

TAHSIK ISLAM SAMPAD

After fixing the component list, I planned the necessary budget with my team mate number 6. Then I went to buy the necessary components with my team number 1,2,4 & 6 from Patuatuli, Puran Dhaka. I also researched about the project feasibility and impacts of the projects while writing the project proposals with my team mate number 3 & 4. Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I contributed in wiring and soldering with my team mate number 2,3,4 & 6.

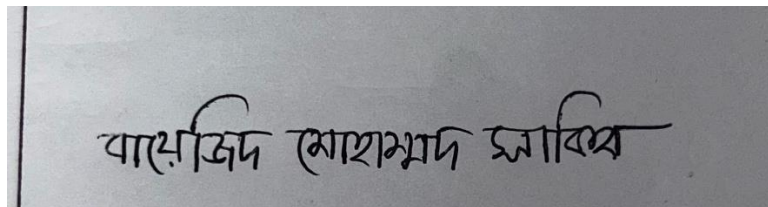
A photograph of a piece of light-colored paper with the word "Sampad" written in a cursive, handwritten style. The signature is dark in color, possibly black ink or a dark marker.

Signature

f. Contribution of team member 6

BAYZID MOHAMMOD SAKIB

After fixing the component list, I planned the necessary budget with my team mate number 6. Then I went to buy the necessary components with my team number 1,2,4 & 6 from Patuatuli, Puran Dhaka. Hardware implementations were done by all of my team members. We sat together while doing our project & got the work done together. Personally, I contributed in wiring and soldering with my team mate number 2,3,4 & 6. I also contributed in the documentation part.



Signature

13.CO PO covered

POs	Program3 Outcomes (POs)	Tic
	Upon graduation, from the Bachelor of Science in Electrical and Electronic Engineering program, the students are expected to have the ability to:	
PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex electrical and electronic engineering problems.	✓
PO2	Problem analysis: Identify, formulate, research literature and analyse complex electrical and electronic engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	✓
PO3	Design/development of solutions: Design solutions for complex electrical and electronic engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓
PO4	Investigation: Conduct investigations of complex electrical and electronic engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	✓
PO5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex electrical and electronic engineering problems, with an understanding of the limitations.	✓
PO6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex electrical and electronic engineering problems.	✓
PO7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts.	✓
PO8	Ethics: Apply ethical principles embedded with religious values, professional ethics and responsibilities, and norms of electrical and electronic engineering practice.	
PO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✓
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	✓
PO11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	✓
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the context of electrical and electronic engineering related technological change.	✓

